DIGITAL LOGIC & COMPUTER DESIGN (DLCD) MINI PROJECT CIRCUIT SUBMISSION

Module Specification & Datapath Circuit Diagram

Selected Module: conv_pe.v with Sequential Q7.8 MAC (mac_q7_8.v)
Project: Hardware Neural Network Accelerator (DLCD Neural Accelerator)
Arithmetic Format: Q7.8 Fixed-Point (16-bit signed 2's complement, Scale = 256)

Team Members & SRN:

Shreyansh Raj PES2UG25CS516	Sarthak Bhowmik PES2UG25CS493
Shaurya Singh PES2UG25CS508	Shreya Anand PES2UG25CS515

1. MODULE FUNCTIONAL ARCHITECTURE & SIGNAL CONTRACT

The Convolutional Processing Element (conv_pe.v) implements a **9-tap weight-stationary, sequential Multiply-Accumulate (MAC) architecture** for 3×3 convolution kernels. It time-multiplexes a single signed Q7.8 multiplier across 9 clock cycles per receptive field, reducing hardware multiplier requirements by **88.89%** compared to parallel spatial unrolling.

Port Name	Dir	Width	Description & Operation	Port Name	Dir	Width	Description & Operation
clk	IN	1 bit	Primary synchronous system clock	clr_acc	IN	1 bit	Asserted at Tap 0 to clear feedback accumulator to 0
reset	IN	1 bit	Synchronous active-high reset (clears counter & valid)	pixel_in	IN	16 bits	Streaming pixel feature input (Q7.8 signed 2's complement)
load_weight	IN	1 bit	Serial shift-enable to pre-load 9 kernel weights	out_acc	OUT	16 bits	16-bit signed accumulated convolution sum (Q7.8 format)
weight_in	IN	16 bits	Serial weight stream fed during configuration	valid_out	OUT	1 bit	Active-high 1-cycle output strobe asserted at Tap 8
mac_en	IN	1 bit	Computation enable; advances tap counter; halts on stall	**Throughput:** 1 valid output feature vector every 9 clock cycles			

2. REGISTER TRANSFER LEVEL (RTL) DATAPATH CIRCUIT SCHEMATIC

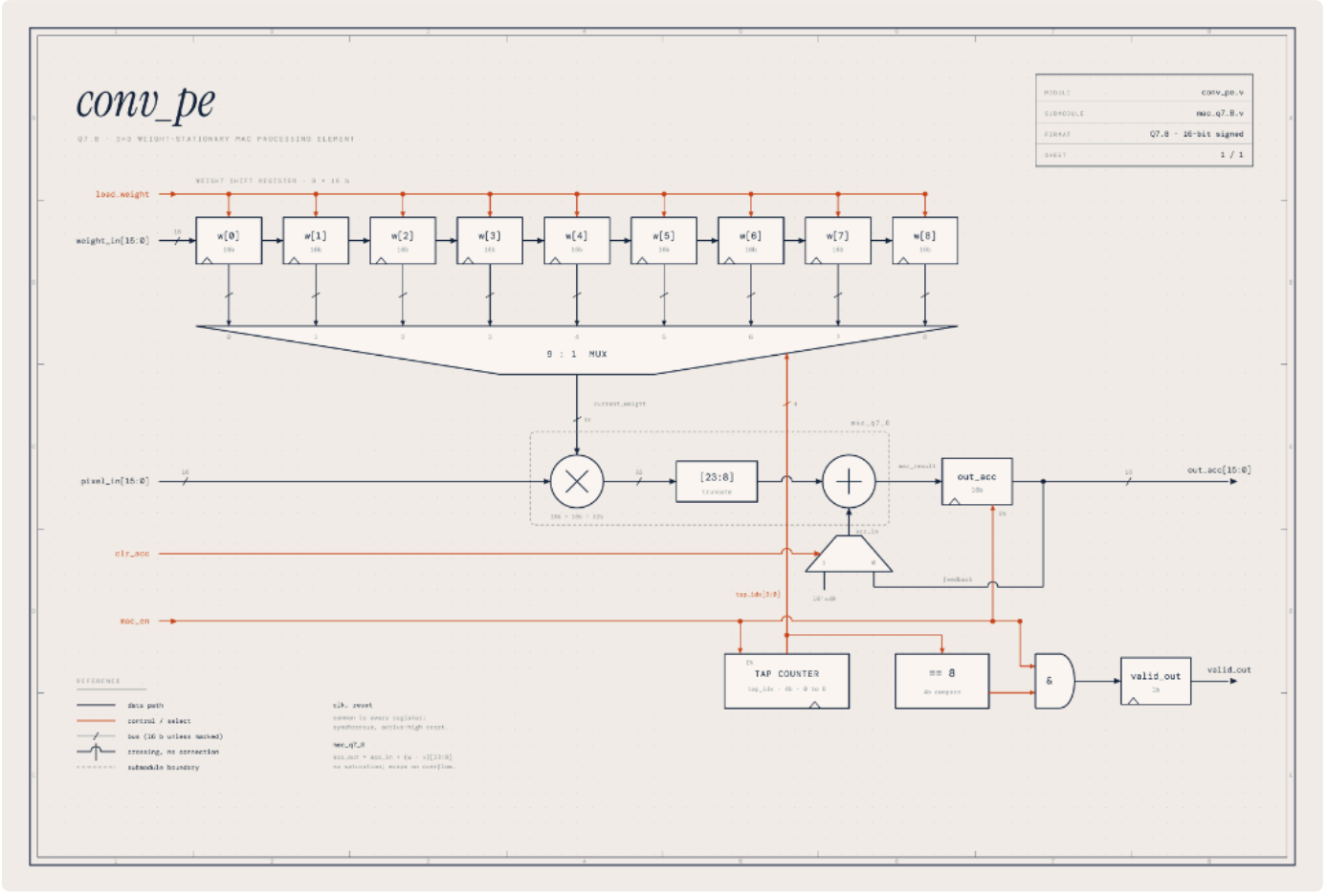


Figure 1: RTL Datapath Architecture Schematic of **conv_pe.v**. Highlights the 9-stage weight shift buffer, 9:1 weight multiplexer, Q7.8 signed multiplier (**mac_q7_8**), bit-slice truncation (**[23:8]**), accumulator adder with synchronous clear MUX, 4-bit Modulo-9 tap counter, and **valid_out** control logic.

3. GATE-LEVEL MICROARCHITECTURE SCHEMATIC

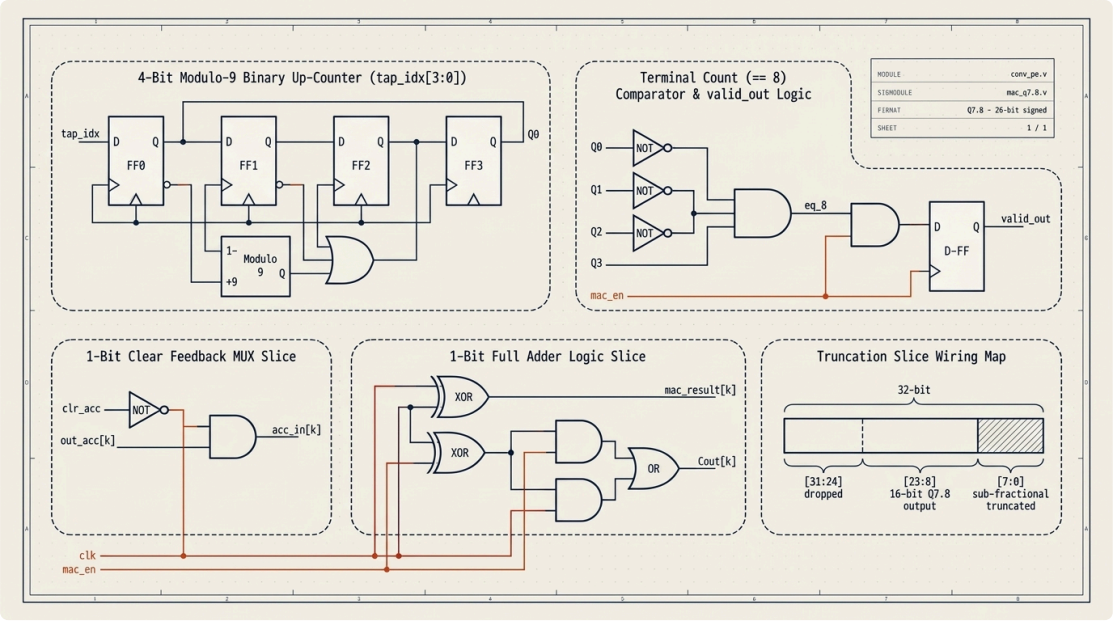


Figure 2: Gate-Level Microarchitecture Implementation. Details the 4-bit synchronous Modulo-9 counter (D-FFs and next-state excitation logic), terminal count comparator (`tap_idx == 8`), `valid_out` gating logic, 1-bit 2:1 multiplexer transmission slice, 1-bit Full Adder logic with carry-chain generation, and Q7.8 bit-slice wiring (`product[23:8]`).

4. SIMULATION WAVEFORM & FUNCTIONAL TIMING VERIFICATION

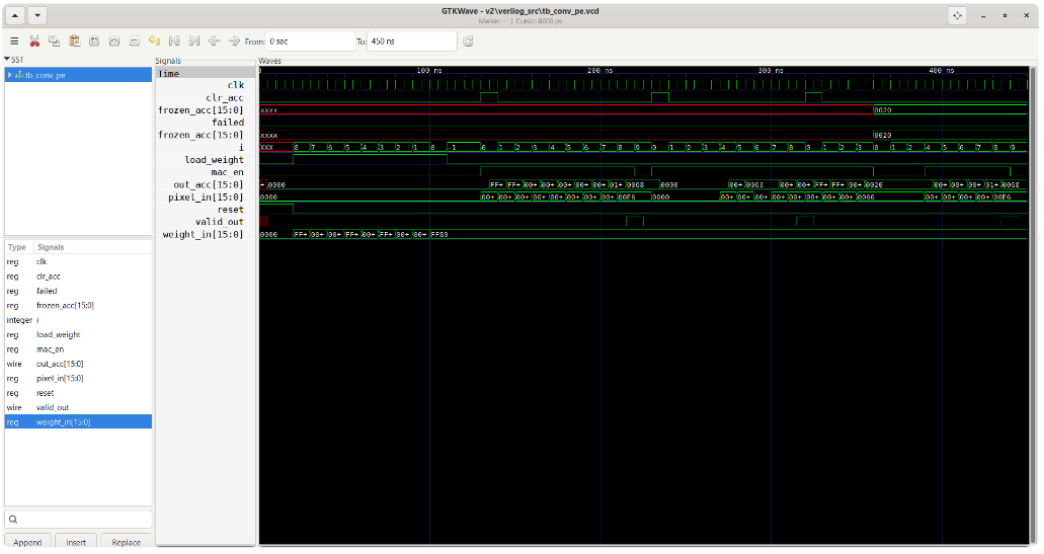


Figure 3: Authentic GTKWave Simulation Trace (`tb_conv_pe.vcd`). Validates: (1) Serial weight loading into internal shift registers, (2) Deterministic 9-cycle accumulation progression (`tap_idx` 0 → 8), (3) Correct assertion of `valid_out = 1` at Tap 8, (4) Mid-stream pipeline stall/resume behavior via `mac_en`, and (5) Hardware regression test status (`failed = 0`).

88.89%	**9 Cycles**	**[-4.9, +8.1]**	**0 Errors (PASSED)**
Multiplier Savings (1 vs 9 DSPs)	Latency per 3×3 Kernel Window	Conv1 Q7.8 Headroom (Range: ±128)	tb_conv_pe Regression Status